Fiscal Year:	FY 2005	**Task Last Updated:**	FY 02/08/2006
PI Name:	Walsworth, Ronald L. Ph.D.		
Project Title:	Investigation of Light Storage in Atomic Ensembles for Space-Based Applications		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Other		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Cambridge	**State:**	MA
Zip Code:	02138	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	08/01/2003	**End Date:**	07/31/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG3 2933		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The recently demonstrated “storage of light” in collective atomic excitations will be applied to investigations of promising methods for the communication of quantum information. The light storage technique is based on reversibly transferring information from a pulse of light to coherently driven atomic media by adiabatically reducing the light pulse group velocity to zero. Methods to be studied include: implementing fast and robust quantum information transfer between light to atoms; entanglement of distant qubits; and novel approaches to scalable quantum networks based on collective excitations. The key feature of these methods is that they are relatively insensitive to temperature, fluctuations in particle distribution, and particle loss. Experimental investigations will be performed to optimize the efficiency of the stored light technique; to minimize decoherence and losses; to study the phase coherence properties of the technique; to demonstrate quantum state transfer between light and atoms; and to use the light storage technique to entangle separated atomic ensembles. The exceptional degree of control possible with quantum optical systems, along with the long		

	coherence times of collective atomic excitations, may enable the application of the stored light technique to the development of a quantum repeater and the realization of quantum-enhanced positioning and timing to meet NASA needs such as improved spacecraft tracking and navigation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Investigations were performed to optimize the efficiency of the photon storage technique; to minimize decoherence and losses; and to demonstrate quantum state transfer between light and atoms. Technical spin-offs of these investigations were also pursued: improved CPT atomic clocks and dynamic optical bistability.
Bibliography Type:	Description: (Last Updated: 07/25/2019)
Articles in Peer-reviewed Journals	Novikova I, Zibrov AS, Phillips DF, Andre A, Walsworth RL. "Dynamic optical bistability in resonantly enhanced Raman generation." Physical Review A. 2004 Jun;69(6):061802. https://doi.org/10.1103/PhysRevA.69.061802 , Jun-2004
Articles in Peer-reviewed Journals	Phillips DE, Novikova I, Wang CYT, Walsworth RL, Crescimanno M. "Modulation-induced frequency shifts in a coherent-population-trapping-based atomic clock." Journal of the Optical Society of America B. 2005 Feb;22 2):305-10. https://doi.org/10.1364/JOSAB.22.000305 , Feb-2005